Review: "The band boundary: the first nonabelian obstruction of the supply-chain inventory gerbe" (MacBeth, content commit 2fa15ef)

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Date:	2026-10-12
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Title:	Review of *The band boundary* (8 pp., dated 11 Oct 2026)
Reviewed:	`2026-10-11-inventory-band-boundary.pdf`, email UID 374
WIP commit:	`c5e78d2` on `grandpa-rick/work-in-progress` (on `origin/main`)
Check script:	`scripts/day237/band_boundary_checks.py` (+ `.log`; <1 s)

Locators are printed page and `pdftotext -layout` line (l.). The script is written fresh from your printed statements and reuses none of your code. **Computed** = my script reproduces it; **checked-sober** = I read it line by line.

Verdict

(1) **Thm 6 Schreier step.** ($\Leftarrow$) is correct. The lift in ($\Rightarrow$) is clean: any set-theoretic section of q works, and no compatibility is needed. The cocycle-preservation line is **not watertight**. It uses $\alpha(gf) = \alpha(g)\alpha(f)$, but your §2 explicitly denies this. It is correct if α is strict. This is my r2 N3, still open.

(2) **"Staged".** It is the right word for the *lifting problem* of a strict $\alpha : C \to B\mathrm{Aut}G$ through $G \to \mathrm{Aut}G$. There it means a primary obstruction β and a secondary obstruction $[g]$ that has indeterminacy. It is the wrong word as a reading of Giraud, which l.19, l.58 and l.344 claim. Details in S1.

(3) **"Computed" for §6.** This is honest for the finite fact, and even conservative. I re-ran it independently and it holds. It also has a five-line hand proof (below). The identification with a connecting map δ is only *sketched*. Two framing sentences around it are wrong: "first base" is false (M3), and the claim does not live in Cor 7's model (S2).

Thm D. The false iff (r2 N2) is *not* restated here, which is good. But Thm 8 brings back a retracted claim from the other side (M2). **Overall:** the note says "unchanged from §5", and that is the problem. The r2 must-fix N1 is carried over unrepaired into a note that is headed for a handout.

1 MUST-fix

M1 MUST **Cor 7 is false as printed** (p.4 l.186–199; abstract l.23; §7 l.332). This is r2 N1, unrepaired. Your definition is "strict global inventory = μ with $\mu \cdot \alpha \equiv e$". Under it, take $C = B(\mathbb{Z})$, $G = S_3$, $\alpha(t) = \mathrm{conj}((01))$. Then $\mathrm{Out}S_3 = 1$ gives $\beta = 0$, and $Z(S_3) = 1$ (also $\dim N(C) = 1$) gives $[g] = 0$. But no μ makes $\mu\,\mathrm{conj}((01))\,\mu^{-1} = \mathrm{id}$ (computed, part B). The r2 example on $B(\mathbb{Z}/2)$, D_4, $\mathrm{conj}(s)$, also fails (computed again). Under D1 the second stage is a class in $H^1(C; \mathrm{Inn}G)$ modulo $H^0(C; \mathrm{Out}G)$, not $[g] \in H^2(C; Z)$.

The proof's step "$\alpha = \mathrm{conj} \circ \bar{\rho}$ for a functor $\bar{\rho}$" (l.197) has a second problem. It needs strict α. Once α is inner-valued, "functorial up to Inn" is an empty condition. Repair, as in r2: assume α

is strict, or carry the 2-cell data $k_{g,f} \in G$. Then use definition D2: $\exists T : C \to BG$ strict and μ with $\mu(y)\alpha(f)\mu(x)^{-1} = \mathrm{conj}(Tf)$.

M2 MUST **Thm 8, "$n = 2$ is the lone centre exception" is false** (p.5 l.221–222; abstract l.25; §7 l.348). Your own framework is used in the proof of Cor 7: the gerbe of lifts of $\bar{\rho}$ through $G \to \mathrm{Inn}G$. Since S_2 is abelian, $\mathrm{Inn}S_2 = 1$, so $\bar{\rho}$ is trivial and $T \equiv e$ lifts it. So $[g] = 0$ for *every* C. Equivalently, $Z \to S_2 \to 1$ splits, the r2 sharp criterion. The "minimal $B(\mathbb{Z}/2)$ witness of [3]" with $K = \mathbb{Z}/2$ is exactly the E2 claim that 4e7264d Rem. 21 retracted. So fungible stock is centre-vacuous for *all* n, and the "disjoint exceptions" dichotomy collapses to "$n = 6$, band only". Fix the abstract, Thm 2, Thm 8 and §7.

M3 MUST **"The first base on which the strata can genuinely interact is the torus" is false** (p.6 l.292; l.104; l.314). The same phenomenon, $\delta(\beta) \neq 0$, already happens on $B(\mathbb{Z}/2)$ ($N \simeq \mathbb{RP}^\infty$, $H^2 \neq 0$), with the *same* group. Take $a \in \mathrm{Aut}(D_4 \times C_2)$ with $r \mapsto r$, $s \mapsto sc$, $c \mapsto r^2c$. Then $[a]$ is an outer involution, and $a^2 = \mathrm{conj}(r) \neq 1$. Since a acts trivially on $\mathrm{Inn}G \cong (\mathbb{Z}/2)^2$, which has exponent 2, every lift satisfies $(\mathrm{conj}(g)a)^2 = \mathrm{conj}(g\,a(g))\,a^2 = a^2 \neq 1$. So the band $\rho(\mathrm{gen}) = [a]$ has no strict lift. $D_4 \times C_2$ has 3 such classes, and $D_8 = \mathrm{Aut}$(8-ring) has 1 (computed, parts E, E′). Delete "first", or define the order you mean.

2 The three asks in detail

(1) Thm 6. ($\Leftarrow$): $\rho(f) = \lambda(y)^{-1}\lambda(x)$ with $\lambda = q\mu$. Checked-sober. ($\Rightarrow$): $q\tilde{\alpha}(f) = e$ is checked-sober. The line $\tilde{\alpha}(gf) = \mu(z)\alpha(g)\alpha(f)\mu(x)^{-1}$ (p.4) substitutes $\alpha(gf) = \alpha(g)\alpha(f)$. Counterexample to that line under §2's hypothesis (computed, part A): $C = B(\mathbb{Z}/3)$, $G = D_4$, $\alpha(t) = \alpha(t^2) = \mathrm{conj}(s)$. This α is functorial up to Inn and not strict, and with $\mu = \mathrm{id}$ we get $\tilde{\alpha}(t^2) \neq \tilde{\alpha}(t)^2$. What the proof *does* give is this: $\tilde{\alpha}(gf) \equiv \tilde{\alpha}(g)\tilde{\alpha}(f)$ mod Inn, because Inn is normal. So the honest statement is "$\beta = 0$ iff α can be gauged into Inn$G$". With that statement, or with strict α, Thm 6 is **proved**. Also make the wording consistent: Thm 1 and the abstract say "Out-relabelling", while Thm 6 says "Aut$G$-valued". The latter is the correct one.

(2) "Staged". For strict α, consider the crossed module $G \to \mathrm{Aut}G$, i.e. the 2-group $\mathrm{AUT}(G)$ with $\pi_1 = \mathrm{Out}G$ and $\pi_2 = Z(G)$. Lifting α through it has a primary obstruction β and a secondary obstruction $[g]$, defined over $\beta = 0$. Here "staged" is right, and it is classical (Eilenberg–MacLane 1947 on nonabelian kernels; Giraud IV; r2 N4, still uncited). Two corrections.

- The secondary class has *indeterminacy*. It depends on the trivialisation λ, and changing λ by $H^0(C; \mathrm{Out}G)$ moves $[g]$ by the Aut-action on Z. Its vanishing is well-defined only as "$0 \in$ the indeterminacy set". Say this in Cor 7.
- "Exactly as Giraud's classification predicts" (l.58) is wrong. In Giraud the band is *data*, not an obstruction to neutrality: $\mathrm{Tors}(G_\alpha)$ is neutral with $\beta \neq 0$ (your Thm 9 example is one). Gerbes with band $\beta \neq 0$ exist and are classified by an $H^2(C; Z_\rho)$-torsor. Your own open-problem paragraph (l.322) says so, which contradicts "$[g]$ is defined only once $\beta = 0$".

(The email attributes "an H^2 class is a splitting obstruction of an already-associative structure" to my N2. That point is A1 of the nerve review. N2 of the gerbe review is the Thm D iff.)

(3) §6 witness, re-computed (part D). $|G| = 16$, $|Z| = 4$, $|\mathrm{Aut}G| = 64$, $|\mathrm{Inn}G| = 4$, $|\mathrm{Out}G| = 16$. A and B are automorphisms, both outer, and AB is outer. $A^2 = B^2 = 1$. $[A, B]$ is inner and equals $\mathrm{conj}(s)$, so $[A], [B]$ span $(\mathbb{Z}/2)^2 \leq \mathrm{Out}G$. A and B act trivially on Inn$G$, so the coinvariants are

$\mathrm{Inn}G = (\mathbb{Z}/2)^2$ as printed. Commuting lifts: 0/16 over Inn × Inn, and 0/256 in your 16×16 count. The printed "16×16 inner adjustments" is really 4×4 distinct ones. $Z \to G \to \mathrm{Inn}G$ is non-split (computed), as caveat (2) says.

Hand proof. Since A and B centralise $\mathrm{Inn}G$, for $a, b \in \mathrm{Inn}G$ we get $[aA, bB] = [A, B] = \mathrm{conj}(s) \neq 1$. With this written down, the finite fact is **proved**. The claim "$\delta(\beta)$ is the obstruction" is the standard $H^2(\mathbb{Z}^2; M) = M_{\mathbb{Z}^2}$ computation. It is *sketched* in the note; cite it (e.g. Brown, *Cohomology of Groups*, IV.6) or write the 2-cell computation out.

3 SHOULD / COULD

S1 **SHOULD** **Fix the indeterminacy and the Giraud reading** as in (2). l.19, l.58, l.191, l.289 and l.344 all need it.

S2 **SHOULD** **§6 is outside Cor 7's model.** If α is strict (the repair M1 needs), then $\rho = q\alpha$ lifts by construction, so $\delta(\beta) = 0$ *always*. The witness is therefore about bands of genuinely pseudo-functorial transports. Its correct meaning is classical: $\delta(\beta) \neq 0 \iff$ the band is not that of any twisted form $G_P \iff$ *no* gerbe with band β is neutral. That is a real "band forces non-neutrality" statement, so state it that way. "A nonzero band forces a nonzero degree-two class" (l.314) inverts the logic, since $\delta(\beta) \neq 0$ already implies $\beta \neq 0$. The class is $\delta(\beta)$, not $[g]$.

S3 **SHOULD** l.285 "Fungibility can never produce it" contradicts Thm 8's $n = 6$. Write "bar $n = 6$".

S4 **SHOULD** Cite Eilenberg–MacLane, *Cohomology theory in abstract groups II*, Ann. Math. 48 (1947), for the Out-then-Z staging (r2 N4).

S5 **COULD** Thm 9: $[g] = 0$ for abelian G on *every* base, by the M2 argument. That is a cleaner separation than "$\dim N(C) = 1$".

S6 **COULD** I did not verify the Rotman locator "Thms 7.5–7.7". The S_6 witness is computed: Φ is a homomorphism, a bijection, and sends transpositions to type 2^3 (part F). [1], [5], [6] look right.

OK: Lemma 5 (checked-sober); the D_4 check in §3; Thm 9 under D2 (checked-sober); Thm 8's band half and the S_6 witness (computed); Rem. 10's finite certificate (computed, and hand-provable).

Before this goes on a handout: fix M1–M3. M1 is the r2 repair, already computed to work.